What Is Beside This Quasar?

An Archival Investigation of J1237+3314

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Independent astronomy project

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Research status. This is an exploratory project report, not a peer-reviewed publication. The central measurements have not yet been independently reproduced. The project does not establish a new gravitational lens or a confirmed dual active galactic nucleus.

1 The question

A known quasar has a faint neighbour in the sky.

A quasar is a very bright galactic nucleus powered by matter accreting onto a supermassive black hole.

Two nearby points of light might be separate, physically associated objects. They might instead be unrelated objects at different distances, or multiple images of one distant source produced by gravitational lensing.

This project began as a search for lensed quasars and developed into a closer examination of the already known pair J1237+3314.

2 The two components

Component A is the bright quasar SDSS J123704.05+331423.8, at redshift approximately 1.288.

Component B is a fainter optical source about $1.62''$ from A. The precise position of the compact radio source JVAS B1234+335 lies very close to B.

Their Gaia DR3 identifiers are:

- A: `1514739204708610432`.
- B: `1514739204711602816`.

The objects and candidate pair were already catalogued. This project does not claim their discovery.

3 What data were examined?

The investigation combines optical positions, radio astrometry and imaging, optical images, and spectroscopy from public astronomical archives.

The data include Gaia, DESI, the DESI Legacy Imaging Surveys, JVAS, FIRST, VLASS, and the Radio Fundamental Catalog.

These observations have different resolutions, apertures, calibrations, and observing dates. A measurement near the pair cannot automatically be assigned to one component.

4 What the analysis reports

The numerical values below are outputs reported by the project. They have not been independently verified by the preparation of this explanatory report.

4.1 The radio position agrees with B

The radio position is approximately 4.7 milliarcseconds from B's Gaia position and approximately $1.619''$ from A.

The reported agreement is consistent with the combined positional uncertainties. It supports associating compact radio emission with B's position.

It does not establish B's distance or prove that it is a second nucleus physically associated with A.

4.2 A spectral feature appears towards B

The DESI spectrum collected through the fibre centred on B contains an emission feature with the reported properties:

- Wavelength: 8528.6 ± 0.4 Å.
- Integrated flux: $(2.78 \pm 0.16) \times 10^{-16}$ erg s^{-1} cm^{-2}.
- Intrinsic full width at half maximum under a single-Gaussian model: 550 ± 30 km s^{-1}.

The quoted errors are statistical fitting estimates, not a complete account of systematic uncertainties. The flux follows the DESI point-source calibration.

The feature is reported in both individual exposures and in an additional extraction from the preprocessed detector images. It is present before correction for contamination from A.

These are internal checks of the same underlying observations, not independent observational confirmation.

4.3 The feature could be oxygen emission

One possible identification is the [O II] doublet, emitted by singly ionized oxygen.

If correct, the gas redshift is approximately 1.2872–1.2879, close to A's redshift.

The profile does not uniquely establish the doublet. Tests of other expected lines provide evidence against some alternative identifications, but other possibilities remain.

Even if the oxygen identification is correct, the gas could belong to the surroundings of A rather than to B.

4.4 Nearby light complicates the spectra

Atmospheric blurring sends light from A into the fibre pointed towards B.

The narrow feature persists over the tested contamination corrections. A corresponding significant feature is not found in the spectrum centred on A.

However, the two spectra were collected at different dates and under different seeing conditions. Their comparison does not precisely locate the gas.

5 What remains unresolved?

The project has not established B's nuclear redshift or confirmed a dual active galactic nucleus.

Possible explanations include:

- A distinct active nucleus physically associated with A.
- Gas around A extending towards B.
- Gas related to an interaction.
- An unrelated radio source projected onto nearby gas.

A compact radio feature connected to a jet from A is another alternative considered in the technical investigation.

The optical and radio brightness patterns motivated a lensing test, but dust, microlensing, source structure, and aperture differences complicate that comparison. Gravitational lensing has not been definitively excluded.

Radio sensitivity thresholds and spatial constraints in the technical analysis are model-dependent supporting results.

6 Verification and reproducibility

The repository should contain one matching release of the scripts, input references, outputs, and reports.

The main documentation files are:

- `release/README.md`.
- `release/data_provenance.md`.
- `release/out/results_table.md`.

A successful execution shows that a workflow runs. It does not automatically validate its assumptions, uncertainty estimates, or physical interpretation.

Repeated runs by the author demonstrate repeatability. Independent reproduction is a separate step.

If execution fails, the error and software environment should be documented rather than hidden.

7 The most useful next step

The immediate priority is independent examination of the spectral feature and radio–optical positional association.

Spatially resolved spectroscopy could test where the gas lies and identify additional lines. Higher-resolution radio imaging could clarify radio emission and source structure.

These observations address different questions and would not automatically confirm a dual AGN.

8 Conclusion

This project re-examines an already known ambiguous pair. It reports a close radio–optical positional association and a spectral feature that may help interpret the system.

The physical relationship remains unresolved.

The purpose of sharing the work is to document the methods, reported evidence, and uncertainties—not to announce a confirmed discovery.

Data credits and AI assistance

Original survey observations retain their providers' terms and acknowledgement requirements. Detailed references and input provenance are recorded in the technical release.

This explanatory report was prepared with assistance from ChatGPT (OpenAI).

Selected data resources

- Gaia: https://gea.esac.esa.int/archive/
- DESI public data: https://data.desi.lbl.gov/public/dr1/
- Legacy Surveys: https://www.legacysurvey.org/
- SIMBAD: https://simbad.cds.unistra.fr/
- VizieR: https://vizier.cds.unistra.fr/

These links identify data services. They do not replace the detailed citations and input references in the technical documentation.